

IgG binders 47 and 48

Full-length parent context for the two domains reported to have affinities comparable to Protein A Z.

9

Binder 47 parents

5

Binder 48 parents

5

Shared parents

Every parent containing binder 48 also contains binder 47 at a different sequence position. Binder 47 has 13 exact sites across nine unique parent sequences; binder 48 has five sites across five parents.

Binder 47 Binder 48 B domain YSIRK LysM Anchor

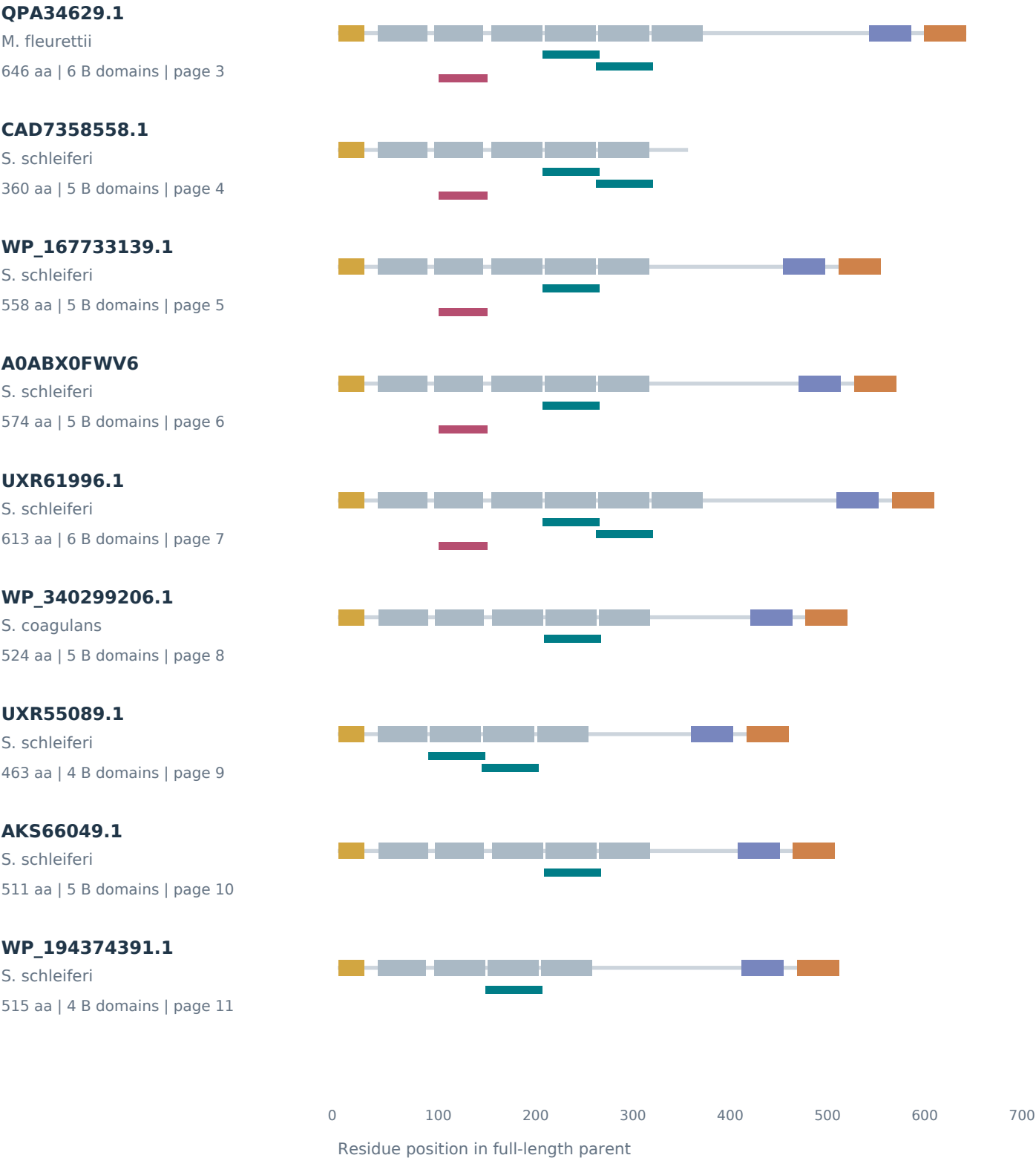
PARENT ACCESSION(S)	LENGTH	BINDERS	PAGE
QPA34629.1 Mammaliicoccus fleurettii	646	47 + 48	3
CAD7358558.1 Staphylococcus schleiferi	360	47 + 48	4
WP_167733139.1 Staphylococcus schleiferi	558	47 + 48	5
A0ABX0FWV6 Staphylococcus schleiferi	574	47 + 48	6
UXR61996.1; WP_262569694.1 Staphylococcus schleiferi	613	47 + 48	7
WP_340299206.1 Staphylococcus coagulans	524	47	8
UXR55089.1; WP_262576444.1 Staphylococcus schleiferi	463	47	9
AKS66049.1 Staphylococcus schleiferi	511	47	10
WP_194374391.1 Staphylococcus schleiferi	515	47	11

Scope: quality-filtered main parent set. Identical whole sequences are merged across accessions. Near matches and excluded records are not included.

Nine parents, one common scale

Gray bars: supported B-family domains. Colored lower tracks: exact binder intervals. Shared parents are shown first.

■ Binder 47 ■ Binder 48 ■ B domain ■ YSIRK ■ LysM ■ Anchor



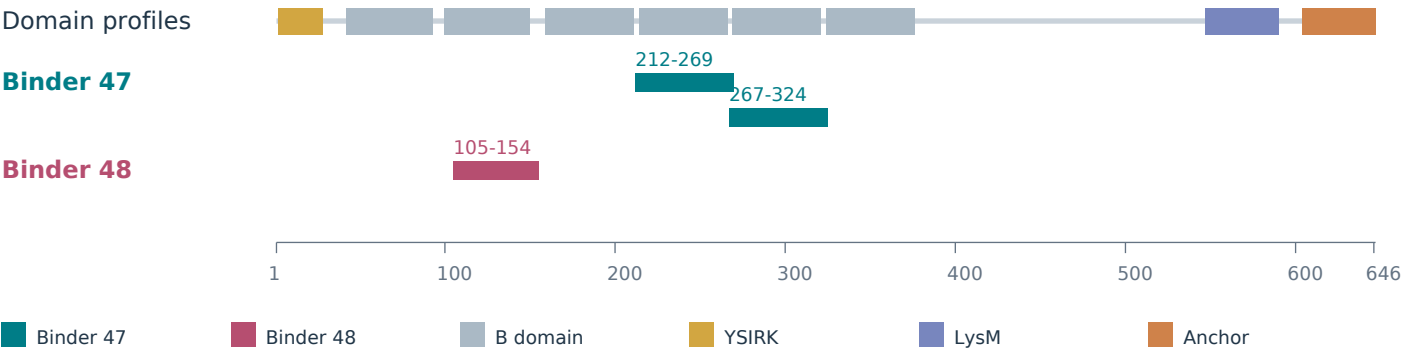
Eight parents are provisionally SpA-like and anchor-supported. CAD7358558.1 has unresolved anchoring context. Profile absence is not proof of biological absence.

QPA34629.1

Mammaliicoccus fleurettii

SHARED PARENT: 47 + 48

Accessions: QPA34629.1
NCBI counterpart(s): QPA34629.1
Length / B-family domains: 646 aa / 6
Provisional context: SpA-like, anchor-supported
Search source(s): TBLASTN search
Database product: B domain-containing protein



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650) B-family domain: 269-320 (PF02216)
B-family domain: 42-92 (PF02216) B-family domain: 324-375 (PF02216)
B-family domain: 100-149 (PF02216) LysM: 547-589 (PF01476)
B-family domain: 159-210 (PF02216) Anchor profile: 604-646 (PF00746)
B-family domain: 214-265 (PF02216)

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

1 MENKNFFSIRKLSIGVGSLIASSLLVSSPTFAEENNANATNTAEQQNAFAEILDLPILNENQRNGFIQSLKDDPSVSK
81 DILAEAKKLNDSSQAGSQPAPDHSADQQNAFYEILHLPNLTEEQRNGFIQSLKDDPSVSADILIQARNVNVAQPTKPAPN
161 VNEDQQNAFYEILNLPNLNEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSSQAAPDYTEAQNAFYEILHLPNLSEDQRNG
241 FIQSLKDDPSVSKDILAEAKKLNDSSQAAPDYTEAQNAFYEILHLPNLSEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSS
321 QAAPDYSEAQQNAFYEILHLPNLTEEQRNGFIQSLKDDPSVSKDILAEAKKLNDSSQAPKKDDGTTNPSDNNNDNQAGQN
401 DNKDDQGNKAEDKGDKNKGDKNKAEDKDNKAEDKGDKNKGDQGNKAEDKGDKNKADKGNKAEDKGDKNKADKGNK
481 AEDKGDKNKADKGNKAEDKGDKNKADQAGKKDNKDMGVAKESGNGNMNSGNKAMPKASTHVVKPGETLNEIAK
561 ANNTTADQIAKDNHLDKNVIMPGQELVINKPAMNENKDAKANSEMKALPETGGESDAALFGTTLAGGLSLAFGSLLLG
641 RSRKSN

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

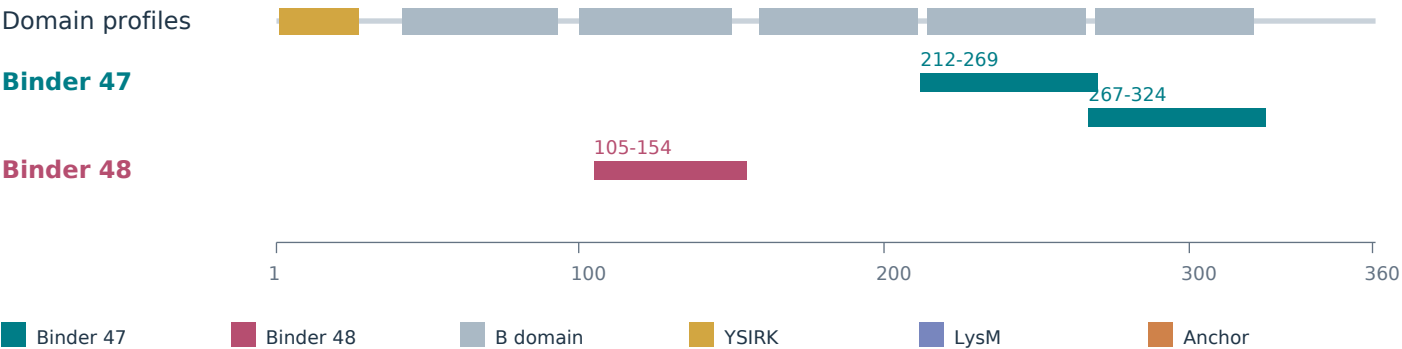
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

CAD7358558.1

Staphylococcus schleiferi

SHARED PARENT: 47 + 48

Accessions: CAD7358558.1
NCBI counterpart(s): CAD7358558.1
Length / B-family domains: 360 aa / 5
Provisional context: B-containing; context unresolved
Search source(s): NCBI protein BLAST; TBLASTN search
Database product: immunoglobulin G binding protein A spa1



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650) B-family domain: 269-320 (PF02216)
B-family domain: 42-92 (PF02216)
B-family domain: 100-149 (PF02216)
B-family domain: 159-210 (PF02216)
B-family domain: 214-265 (PF02216)

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

1 MENKNFFSIRKLSIGVGSLIASSLLVSSPTFAEENNANATNTAEAQNAFAEILDLPNLNENQRNGFIQSLKDDPSVSK
81 DILAEAKKLND SQAGSQPADHSA**DQQNAFYEILHLPNL TEEQRNGFIQSLKDDPSVSADILIQARNVNVQA**PTKPAPN
161 VNEDQQNAFYEILNLPNLNEDQRNGFIQSLKDDPSVSKDILAEAKKLND**SQAAPDYTEAQNAFYEILHLPNL**SEDQRNG
241 **FIQSLKDDPSVSKDILAEAKKLND**SQAAPDYTEAQNAFYEILHLPNL**SEDQRNGFIQSLKDDPSVSKDILAEAKKLND**
321 **QAAPDY**SEAQNAFYEILHLLTLLKNNVMVSSKALKMTQA

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed. This record passes the completeness filter, but anchoring context remains unresolved.

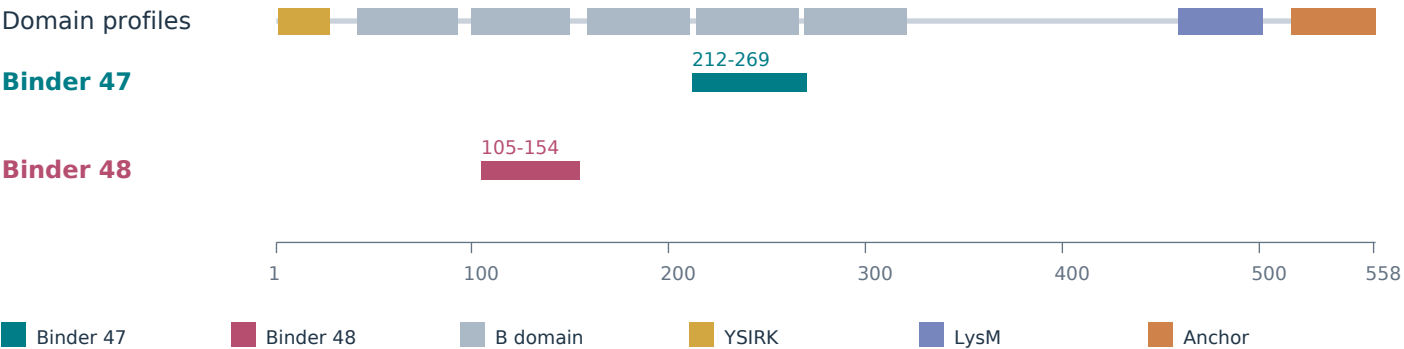
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

WP_167733139.1

Staphylococcus schleiferi

SHARED PARENT: 47 + 48

Accessions: WP_167733139.1
NCBI counterpart(s): WP_167733139.1
Length / B-family domains: 558 aa / 5
Provisional context: SpA-like, anchor-supported
Search source(s): NCBI protein BLAST
Database product: staphylococcal protein A



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650)	B-family domain: 269-320 (PF02216)
B-family domain: 42-92 (PF02216)	LysM: 459-501 (PF01476)
B-family domain: 100-149 (PF02216)	Anchor profile: 516-558 (PF00746)
B-family domain: 159-210 (PF02216)	
B-family domain: 214-265 (PF02216)	

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

```
1  MENKNFFSIRKLSIGVGSLIASSLLVSSPTFAEENNANATNTAEQQNAFAEILDLPNLNENQRNGFIQSLKDDPSVSK
81  DILAEAKKLNDSDAGSQPAPDHSADQQNAFYEILHLPNL TEEQRNGFIQSLKDDPSVSADILIQARNVNVAQPTKPAPN
161 VNEDQQNAFYEILNLPNLNEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSQAAPDYTEAQNAFYEILHLPNLSE
241 FIQSLKDDPSVSKDILAEAKKLNDSQAAPDYSEAQNAFYEILHLPNL TEEQRNGFIQSLKDDPSVSKEILAEAKKLND
321 QAPKKDDGTTNPSDNNNNNDQAGQNDNKDDQGNKAEDKGDKNKADKGNKAEDKGDKNKADKGHAEDKGD
401 NKADKGHAEDKGDGVDKGNKADQAGKDNKDMGVAKEQSGNNGMNNSGNKAMPKASTHVVKPGETLNEIAKANNTTADQ
481 IAKDNHLADKNVIMPGQELVINKPAMMENKDAKANSEMKALPETGGESDAALFGTTLAGGLSLAFGSLLLGRSRKSN
```

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

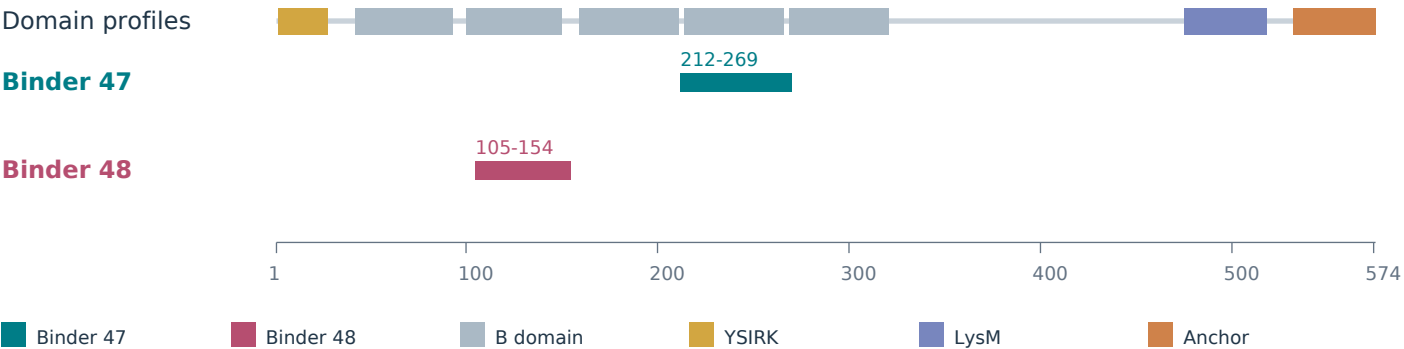
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

A0ABX0FWV6

Staphylococcus schleiferi

SHARED PARENT: 47 + 48

Accessions: A0ABX0FWV6
NCBI counterpart(s): NHA33140.1
Length / B-family domains: 574 aa / 5
Provisional context: SpA-like, anchor-supported
Search source(s): Pfam search
Database product: LysM domain-containing protein



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650)	B-family domain: 269-320 (PF02216)
B-family domain: 42-92 (PF02216)	LysM: 475-517 (PF01476)
B-family domain: 100-149 (PF02216)	Anchor profile: 532-574 (PF00746)
B-family domain: 159-210 (PF02216)	
B-family domain: 214-265 (PF02216)	

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

```
1  MENKNFFSIRKLSIGVGSLIASSLLVSSPTFAEENNANATNTAEQQNAFAEILDLPNLNENQRNGFIQSLKDDPSVSK
81  DILAEAKKLNDSQAGSQPAPDHSADQQNAFYEILHLPNL TEEQRNGFIQSLKDDPSVSADILIQARNVNVAQPTKPAPN
161 VNEDQQNAFYEILNLPNLNEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSQAAPDYTEAQNAFYEILHLPNLSEDQRNG
241 FIQSLKDDPSVSKDILAEAKKLNDSQAAPDYSEAQNAFYEILHLPNL TEEQRNGFIQSLKDDPSVSKEILAEAKKLNDS
321 QAPKKDDGTTNPSDNNNDNQAGQNDNKDDQGNKAEDKGDKDNKGDQGNKAEDKGDKDNKGDQGNKAEDKGDKDNKADKG
401 NKAEDKGDKDNKADKNKADKGHKAEDKGDKVKGKADQAGKDNKDMGVAKEQSGNGMNNSGNKAMPKASTHVVKPG
481 ETLNEIAKANNTTADQIAKDNHLADKNVIMPGQELVINKPAMMENKDAKANSEMKALPETGGESDAALFGTTLAGGLSL
561 AFGSLLLGRSRKSN
```

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

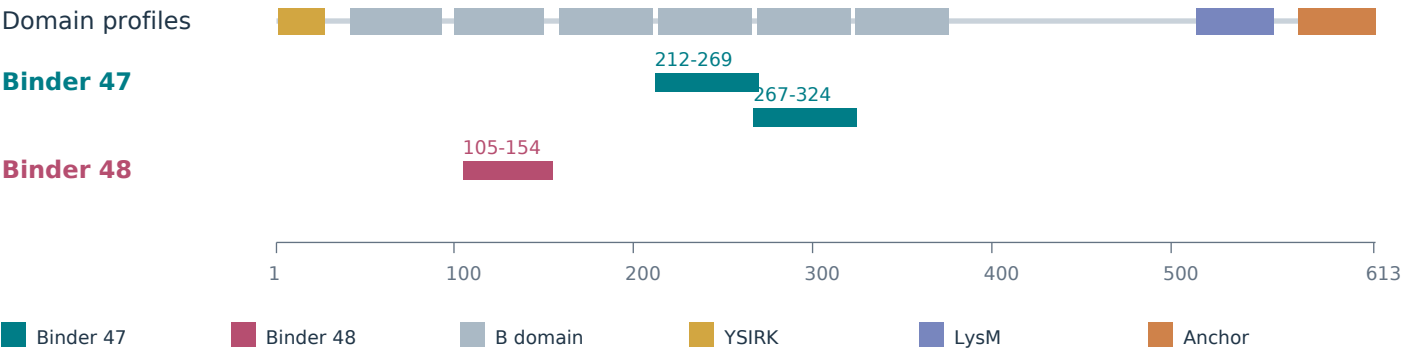
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

UXR61996.1

Staphylococcus schleiferi

SHARED PARENT: 47 + 48

Accessions: UXR61996.1; WP_262569694.1
NCBI counterpart(s): UXR61996.1; WP_262569694.1
Length / B-family domains: 613 aa / 6
Provisional context: SpA-like, anchor-supported
Search source(s): NCBI protein BLAST; TBLASTN search
Database product: B domain-containing protein



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650) B-family domain: 269-320 (PF02216)
B-family domain: 42-92 (PF02216) B-family domain: 324-375 (PF02216)
B-family domain: 100-149 (PF02216) LysM: 514-556 (PF01476)
B-family domain: 159-210 (PF02216) Anchor profile: 571-613 (PF00746)
B-family domain: 214-265 (PF02216)

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

1 MENKNFFSIRKLSIGVGSLIASSLLVSSPTFAEENNANATNTAEQQNAFAEILDLPNLNENQRNGFIQSLKDDPSVSK
81 DILAEAKKLNDSSQAGSQPAPDHSADQQNAFYEILHLPNLTEEQRNGFIQSLKDDPSVSADILIQARNVNVAQPTKPAPN
161 VNEDQQNAFYEILNLPNLNEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSSQAAPDYTEAQNAFYEILHLPNLSEDQRNG
241 FIQSLKDDPSVSKDILAEAKKLNDSSQAAPDYTEAQNAFYEILHLPNLSEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSS
321 QAAPDYSEAQQNAFYEILHLPNLTEEQRNGFIQSLKDDPSVSKEILAEAKKLNDSSQAPKKDDGTTNPSDNNNNNDNQAGQN
401 DNKDDQGNKAEDKGDKNKGQDNKAEDKGDKNKGQGNKAEDKGDKNKGQGNKAEDKGDKNKADKGNKADQA
481 GKKNKDMGVAKEQSGNNGMNSGNKAMPKASTHVVKPGETLNEIAKANNTTADQIAKDNHLADKNVIMPGQELVINKPA
561 MMNENKDAKANSEMKALPETGGESDAALFGTTLAGGLSLAFGSLLLGRSRKSN

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

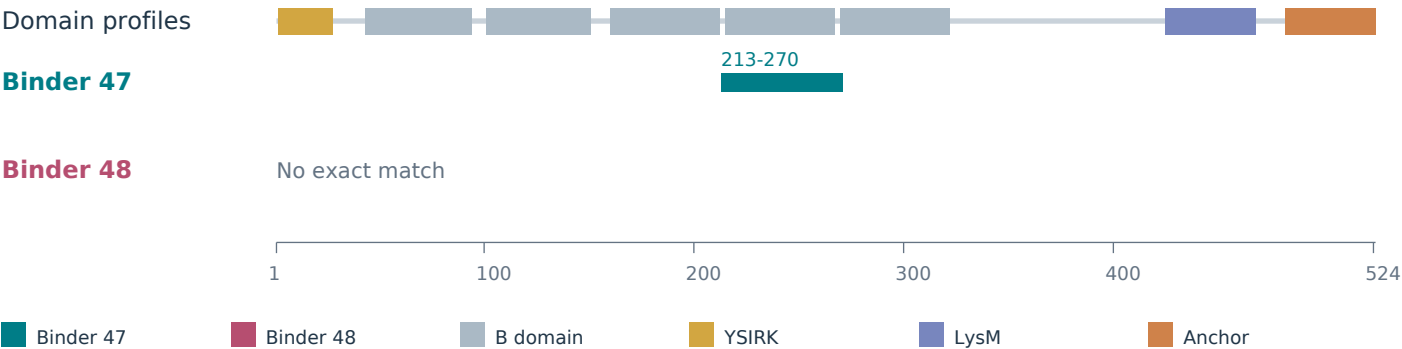
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

WP_340299206.1

Staphylococcus coagulans

BINDER 47 PARENT

Accessions: WP_340299206.1
NCBI counterpart(s): WP_340299206.1
Length / B-family domains: 524 aa / 5
Provisional context: SpA-like, anchor-supported
Search source(s): NCBI protein BLAST
Database product: staphylococcal protein A



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650) B-family domain: 270-321 (PF02216)
B-family domain: 43-93 (PF02216) LysM: 425-467 (PF01476)
B-family domain: 101-150 (PF02216) Anchor profile: 482-524 (PF00746)
B-family domain: 160-211 (PF02216)
B-family domain: 215-266 (PF02216)

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

```
1  MENKNFFSIRKLSIGVGSCLIASSLLVSSPTFAEENNNANATNTAEAQNAFAEVLNLPNLNEDQRNGFIQSLKDDPSAS
81  KDVLAEAQKLND SQAGSQPAPDHSADQQNAFY EILHLPNL TEEQRNGYIQSLKDDPSVSADILIQAKNVNVAQAPTQPAP
161 NVNEDQQNAFY EILNLPNLNEDQRNGFIQSLKDDPSVSKDVLAEAKKLND SQAAPDYTEAQNAFY EILHLPNLSEDRN
241 GFIQSLKDDPSVSKDILAEAKKLND SQAAPDYSEAQNAFY EILHLPNL TEEQRNGFIQSLKDDPSVSKDILAEAKKLND
321 SQAPKKDDGTTNPSDNNNNNDNQAGQNDNKGQDNKGQGNKAEDKGGKADKGNKAEDKGGKADQGHKADQAGKKDNKDMG
401 AAKEQSGNNGMNSGKAMPKASTHVVKPGETLNEIAKANNTTADQIAKDNHLADKNVIMPGQELVINKPAMMNENKDAK
481 ANSEM KALPETGGESDAALFGTTLAGGLSLAFGSLLLGRSRKSN
```

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

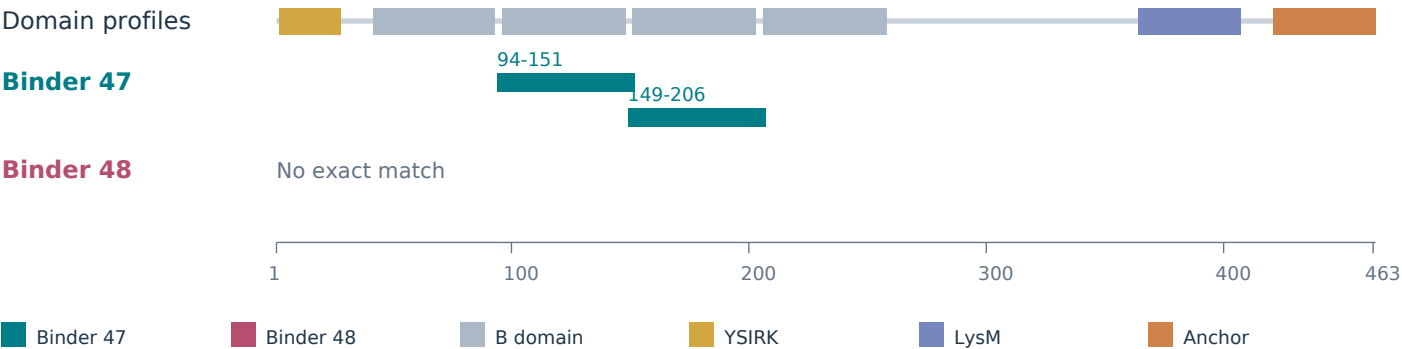
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

UXR55089.1

Staphylococcus schleiferi

BINDER 47 PARENT

Accessions: UXR55089.1; WP_262576444.1
NCBI counterpart(s): UXR55089.1; WP_262576444.1
Length / B-family domains: 463 aa / 4
Provisional context: SpA-like, anchor-supported
Search source(s): NCBI protein BLAST; TBLASTN search
Database product: B domain-containing protein



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650) LysM: 364-406 (PF01476)
B-family domain: 42-92 (PF02216) Anchor profile: 421-463 (PF00746)
B-family domain: 96-147 (PF02216)
B-family domain: 151-202 (PF02216)
B-family domain: 206-257 (PF02216)

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

```
1  MENKNFFSIRKLSIGVGSLIASSLLVSSPTFAEENNANATNTAEAQNAFAEILDLPNLNENQRNGFIQSLKDDPSVSK
81  DILAEAKKLNDQAAPDYTEAQNAFYIEILHLPNLSEDQRNGFIQSLKDDPSVSKDILAEAKKLNDQAAPDYTEAQNA
161 FYEILHLPNLSEDQRNGFIQSLKDDPSVSKDILAEAKKLNDQAAPDYSEAQNAFYIEILHLPNLTEEQRNGFIQSLKDD
241 PSVSKEILAEAKKLNDQAAPKDDGTTNPSDNNNNNDQAGQNDNKDDQGNKAEDKGDKNKGDQDNKAEDKGDKDDKDNK
321 ADKGNKADQAGKDNKDMGVAKEQSGNNGMNNNSGNKAMPKASTHVVKPGETLNEIAKANNTTADQIAKDNHLADKNVIMP
401 GQELVINKPAMMENKDAKANSEMKALPETGGESDAALFGTTLAGGLSLAFGSLLLGRSRKSN
```

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

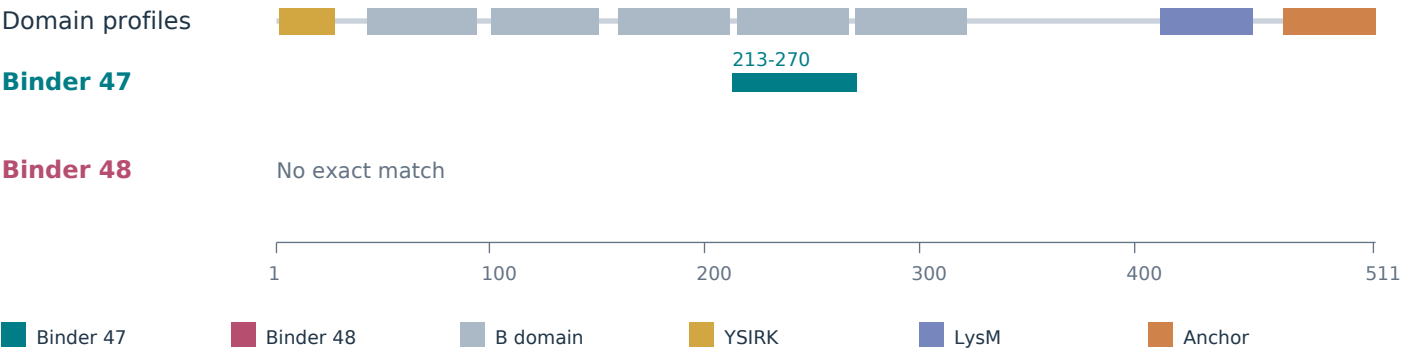
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

AKS66049.1

Staphylococcus schleiferi

BINDER 47 PARENT

Accessions: AKS66049.1
NCBI counterpart(s): AKS66049.1
Length / B-family domains: 511 aa / 5
Provisional context: SpA-like, anchor-supported
Search source(s): TBLASTN search
Database product: peptidoglycan-binding protein LysM



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650) B-family domain: 270-321 (PF02216)
B-family domain: 43-93 (PF02216) LysM: 412-454 (PF01476)
B-family domain: 101-150 (PF02216) Anchor profile: 469-511 (PF00746)
B-family domain: 160-211 (PF02216)
B-family domain: 215-266 (PF02216)

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

```
1  MENKNFFSIRKLSIGVGSLIASSLLVSSPTFAEENNNANATNTAEAQNAFAEVLNLPNLNEDQRNGFIQSLKDDPSAS
81  KDVLAEAQKLNDQAGSQPAPDHSADQNAFYIEILHLPNLTEEQRNGYIQSLKDDPSVSADILIAKNVNVQAAPTQPAP
161 NVNEDQQNAFYIEILNLPNLNEDQRNGFIQSLKDDPSVSKDVLAEAKKLNDQAAAPDYTEAQNAFYIEILHLPNLSEDQRN
241 GFIQSLKDDPSVSKDILAEAKKLNDQAAAPDYSEAQNAFYIEILHLPNLTEEQRNGFIQSLKDDPSVSKDILAEAKKLND
321 SQAPKKDDGTTNPSDNNNNNDNAGQNDNKGQDNKGQGNKAEDKGGKADQGHKADQAGKKDNKDMGAAKEQSGNNGMNN
401 SGNKAMPKASTHVVKPGETLNEIAKANNTTADQIAKNHLADKNVIMPGQELVINKPAMMNENKDAKANSEMKALPETGG
481 ESDAALFGTTLAGGLSLAFGSLLLGRSRKSN
```

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

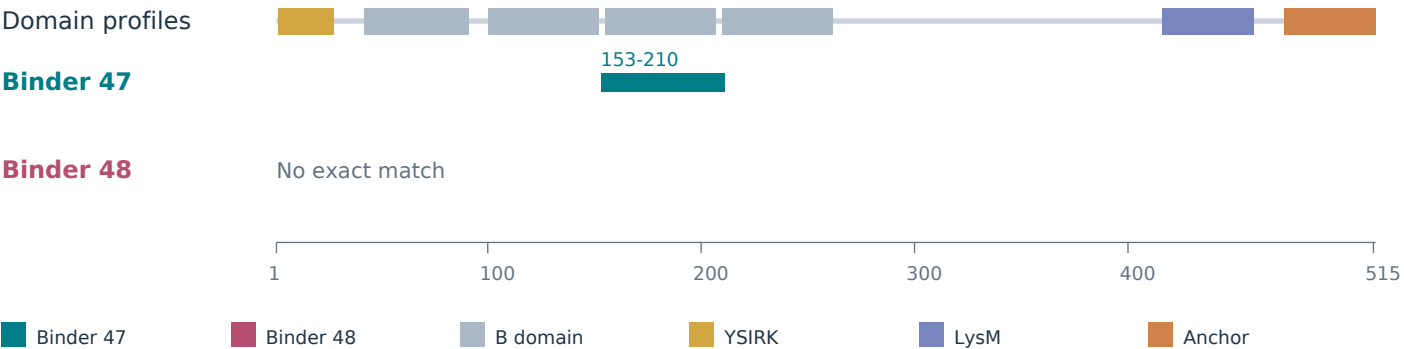
Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.

WP_194374391.1

Staphylococcus schleiferi

BINDER 47 PARENT

Accessions: WP_194374391.1
NCBI counterpart(s): WP_194374391.1
Length / B-family domains: 515 aa / 4
Provisional context: SpA-like, anchor-supported
Search source(s): NCBI protein BLAST
Database product: B domain-containing protein



ANNOTATED DOMAIN COORDINATES

YSIRK signal profile: 2-27 (PF04650) LysM: 416-458 (PF01476)
B-family domain: 42-90 (PF02216) Anchor profile: 473-515 (PF00746)
B-family domain: 100-151 (PF02216)
B-family domain: 155-206 (PF02216)
B-family domain: 210-261 (PF02216)

FULL PARENT SEQUENCE

Teal = binder 47 residues; rose = binder 48 residues. All matches are 100% identical.

```
1  MENKNFFSIRKLSIGVGSLIASLLVSSPTFAEENNANATNTAEAQNAFAEILDLPNLNENQRNGFIQSLKDDPSVSA
81  DILIQARNVNVAQAPTKPAPNVNEDQQNAFYEILNLPNLNEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSSQAAPDYTEA
161 QQNAFYEILHLPNLSEDQRNGFIQSLKDDPSVSKDILAEAKKLNDSSQAAPDYSEAQQNAFYEILHLPNLTEEQRNGFIQS
241 LKDDPSVSKDILAEAKKLNDSSQAPKKDDGTTNPSDNNNNNDNQAGQNDNKDDQGKNAEDKGDKNKGDQDNKAEDKGDKN
321 KGDQGNKAEDKGDKDDKNKADKGNKAEDKGDKDDKNKADKGHKAEDKDGKVDKGNKADQAGKKDNKDMGVAKEQSGNN
401 GMNNSGNKAMPKASTHVVKPGETLNEIAKANNTTADQIAKDNHLADKNVIMPGQELVINKPAMMENKDAKANSEMKALP
481 ETGGESDAALFGTTLAGGLSLAFGSLLLGRSRKSN
```

Domain rectangles show profile alignment coordinates, not experimentally measured boundaries. A targeted 38-model Pfam scan was used; dedicated signal-peptide/TM prediction was not performed.

Experimental evidence applies to the isolated binder sequences. Full-length binding, localization and a unique historical source are not established by these exact matches.